UNIVERSITI TEKNOLOGI MALAYSIA

ASSIGNMENT 1 (15%)

SEMESTER II 2021/2022

CODE OF SUBJECT :

NAME OF SUBJECT :**SPECIAL TOPIC OF SOFTWARE ENGINEERING**

YEAR / COURSE : **JOHOR BAHRU**

ASSIGNMENTS: Cloud Computing: Planning For The Large-Scale System

DELIVERABLES:

1. Technical Reports
2. Due Date: 3 June 2022
3. Do in group

OBJECTIVES :

- Analyze existing network design to solve cloud computing issues
- Analyze and design the proposed solution by using the AWS solution.
- Evaluate and compare the network design between the on-premises design and AWS cloud computing.

INSTRUCTION AND QUESTION:

1. Identify the given case study in the Attachment.

2. From the identified case study, as a Solution Architect, perform the following tasks:
   i. Discuss the case study in detail as the MediHealth Sdn Bhd decided to move from the on-premises network to the cloud solution.
      a. Subsystems
      b. Costings
      c. Advantages of AWS

3. Design Solution: Develop your design solution using the following requirements:
   i. Costing
   ii. Cloud security; shared responsibility model, IAM, data security
   iii. Identify VPC, Availability Zones, and EC2

4. Prepare the following submission:
   a. **Technical Paper**. Prepare a technical report to follow IEEE Conference format (minimum 8 pages) with the following structure:
      i. Abstract.
      ii. Introduction - Describe the problems of the on-premises network design and relate them to the given case study.
      iii. Related Works - Discuss the state-of-arts or review how AWS can solve the on-premises network design.
      iv. Backgrounds – Compare the existing network design and AWS network architecture (give an example of an existing industry at least THREE (3) that already implemented AWS) to justify your selection of the AWS implementation.
      v. The proposed solution - The implementation of the AWS based on the given requirements (3 (i-iii)) and support with the network diagram.
      vi. Result and Discussion – Compare and evaluate the implementation of the AWS to the given case study
      vii. Conclusion
      viii. References.
   b. **Final Submission**
      i. Technical Paper

Attachment

Case Study: (HealthNet)

Overview

This case study explained the healthcare-related supply business and the development of the HealthNet (medical supply network). In 2020, MediHealth Sdn. Bhd. was planning to create a medical supply information network to react to threats from the competitors from the US companies. From a small company to supply medical health, which is established in 1998, MediHealth Sdn. Bhd. has grown to become the largest medical-related supplier in Malaysia and this company makes around 20 percent of its revenue from medical devices and medicine totalling RM20 million in 2020

HealthNet Infrastructure

The massive order of medical devices and medicine in recent years now requires a high level of information technology support to ensure shipments are correctly assigned and that they arrive at the final destination on time. All this is handled by **MediHealth Ordering System (MOS), MediHealth Delivery System (MDS**), and **Information for Pharmaceutical System (IPS).**

Since 1999, IPS has become an integral part of MediHealth's integration system. Each ordering bill can be booked on the selected date and sector with instant confirmation. Once booked, the information of the medical device and pharmaceutical was recorded in real-time, starting from companies' stock until shipped to the destination (Hospital/clinic). Delivery companies that are appointed delivery agents (DHL) can immediately determine where their shipments are at any stage. Medical supply details can be forwarded prior to departure for pre-arrival customs clearance at the destination.

In 2001 MediHealth joined forces with another top 3 medical supplier companies to establish a Medical Device and Pharmaceutical Supply System known as **HEALTHNET**, developed to provide a fast and accurate medical device and pharmaceutical information to all hospitals and clinics in Malaysia. By 2020 the total number of hospitals and clinics connected by HEALTHNET amounted to almost all public and private hospitals in Malaysia.

There are currently 100 IPS display units in 50 sites around the MediHealth network. Since May 2003, 10 delivery agents in 500 sites have related to IPS through HEALTHNET, enabling them to make direct booking entries and gain instant access to medical devices and pharmaceutical information. Enhancements to the system since its inauguration have resulted in instant and automated printing of ordering bills, production of sales reports, and generation of management data. In addition, the web portal for Healthnet has been developed to provide inquiry services for medical devices and pharmaceuticals in order to make ordering and delivery monitoring faster and in real-time. The hospitals/clinics can also obtain online shipment status via ordering status via delivery agent API (DHL) through HealthNet Network.

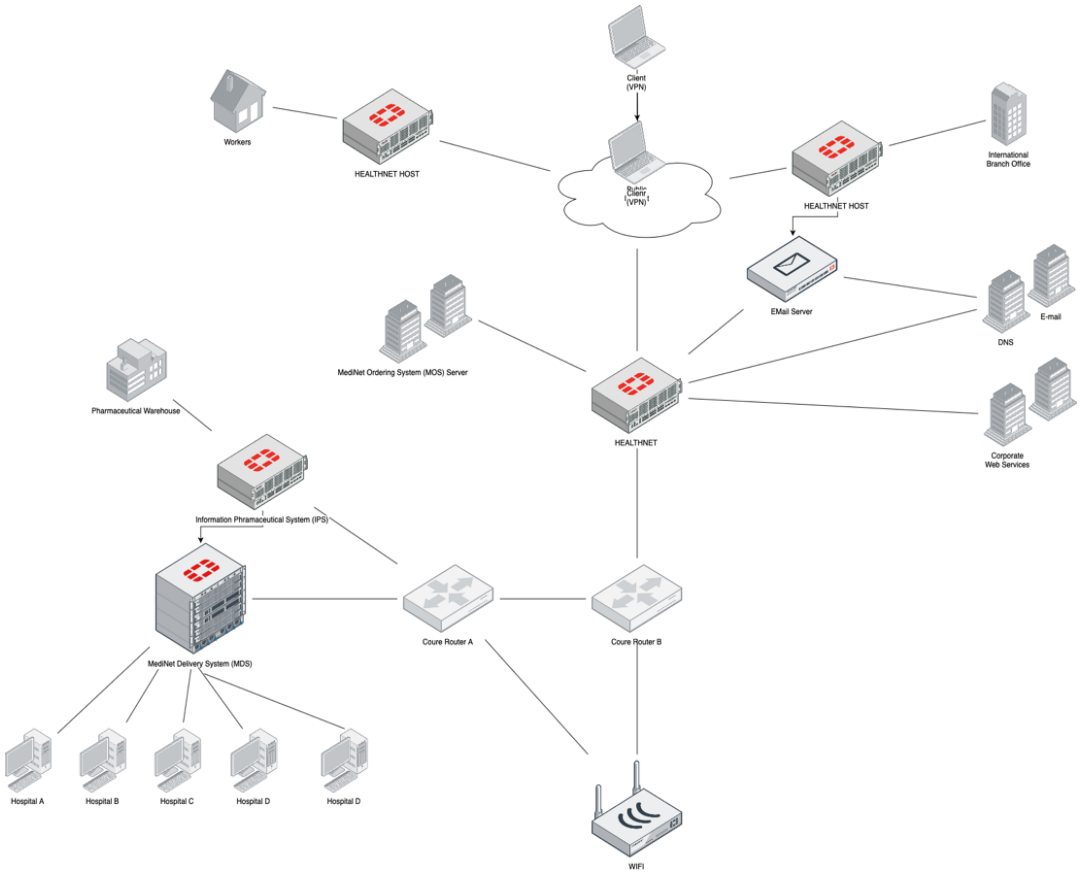


Figure 1: Healthnet Network